Experiment:8

Aim: CI/CD Pipeline to host a website on AWS.

LO4, LO6 mapped:

LO4:: Containerize, orchestrate, and deploy applications using Docker, Docker Compose, Kubernetes, and cloud infrastructure tools.

LO6: Develop and demonstrate an end-to-end DevOps pipeline mini-project integrating all stages of DevOps from code commit to deployment and monitoring.

Theory:

AWS CI/CD Pipeline is an automated software delivery process that continuously integrates code changes, builds the application, and deploys the updated application to a target environment. CI/CD stands for Continuous Integration and Continuous Delivery/Deployment. It reduces manual effort and helps deliver software changes quickly and consistently.

In this experiment, GitHub is used as the source-code repository. Whenever changes are made to the repository, AWS CodePipeline can automatically detect the changes and start the pipeline. CodePipeline coordinates the different stages of the software delivery process.

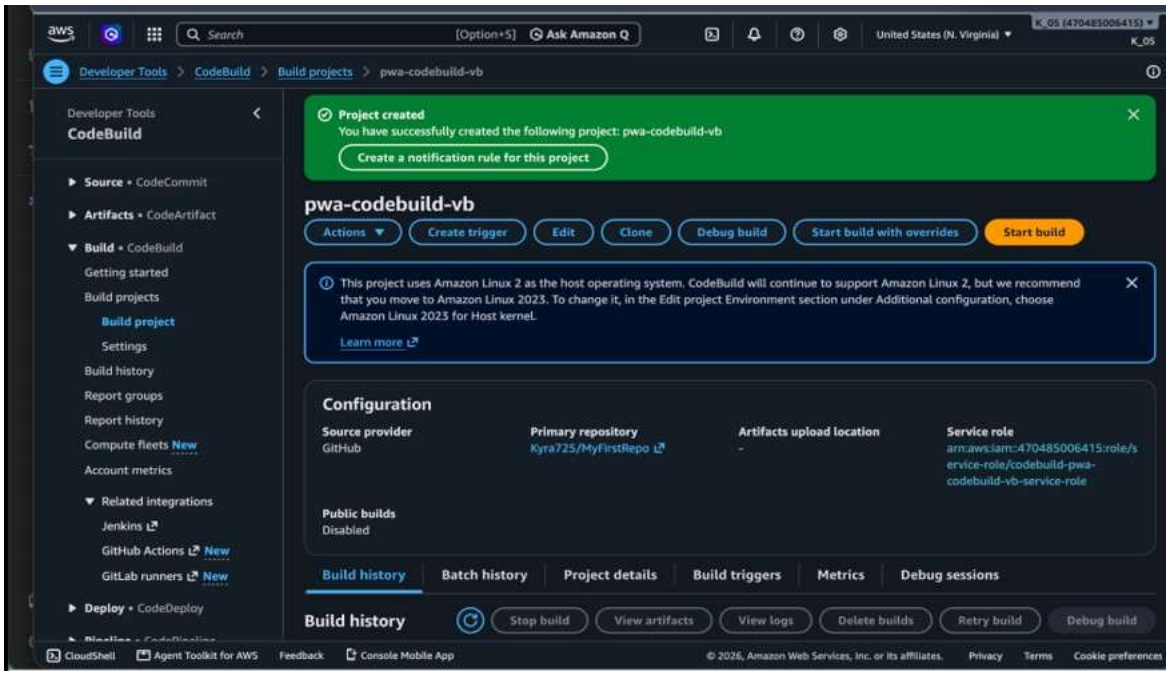
AWS CodeBuild is used as the build stage. It retrieves the source code and executes the commands specified in the buildspec.yml file. The buildspec file defines the commands and artifacts required during the build process.

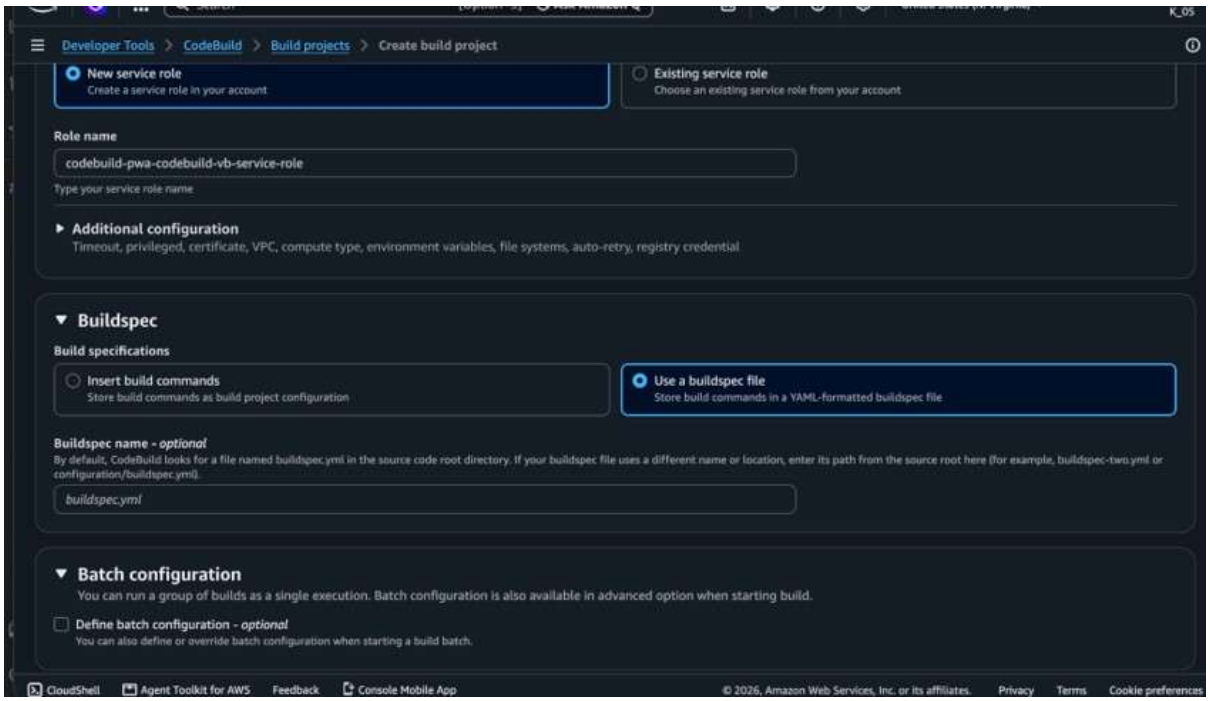
Amazon S3 is used to store the build artifacts and deploy the website. The deployed files can be hosted as a static website using S3 Static Website Hosting. An index.html file is specified as the index document, allowing the website to be accessed through the S3 website endpoint. The complete workflow implemented in this experiment is:

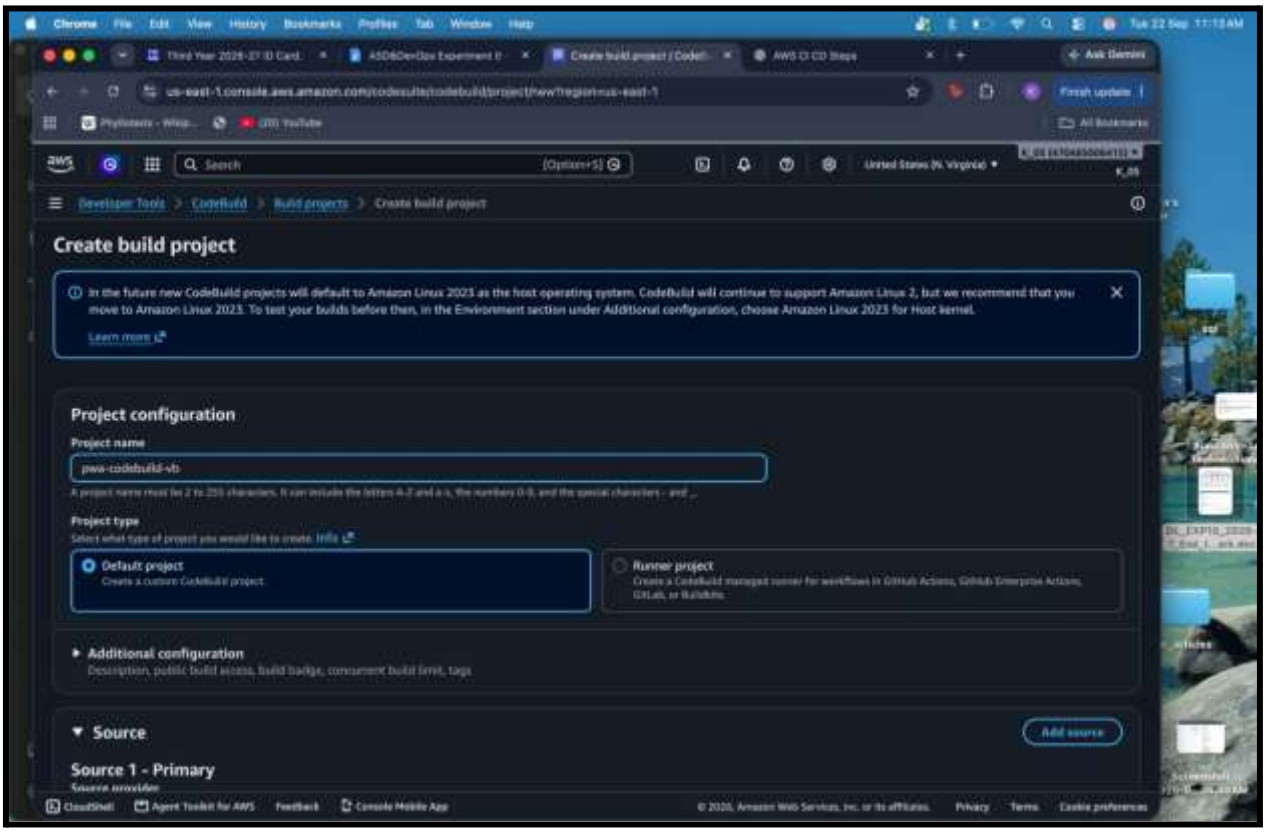
GitHub → AWS CodePipeline → AWS CodeBuild → Amazon S3 → Static Website

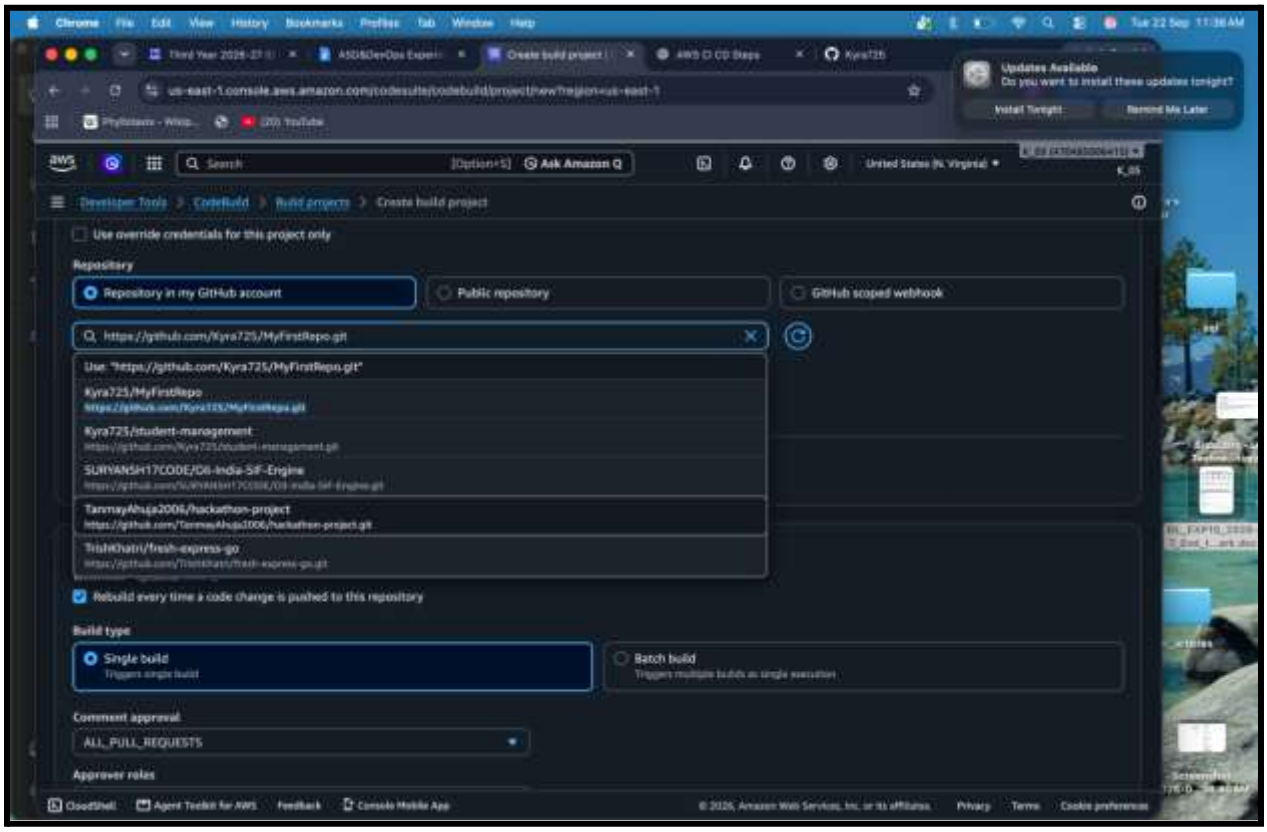
When a change is made to the source code, CodePipeline automatically triggers the required stages and updates the deployed website. This demonstrates automated deployment and continuous delivery.

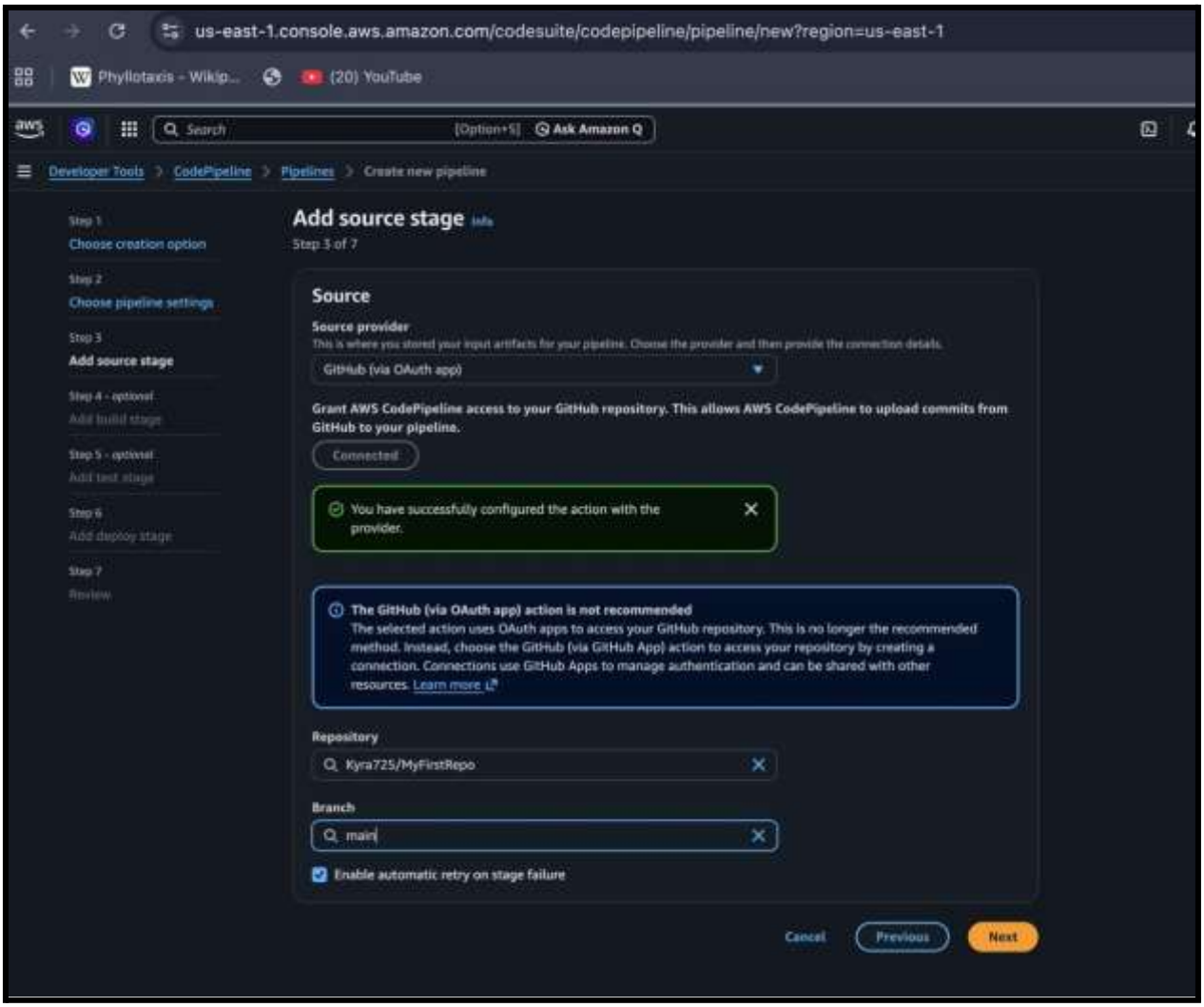
Output:

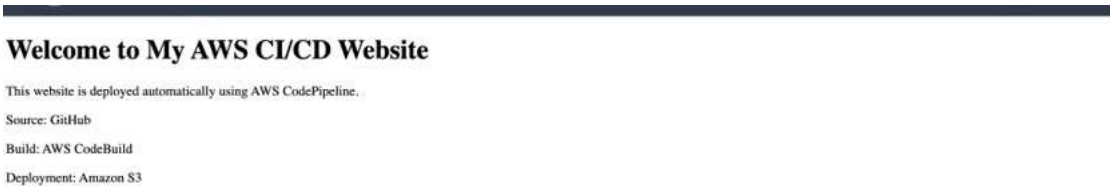

Developer Tools > CodeBuild > Build projects > pwa-codebuild-vb
Project created
You have successfully created the following project: pwa-codebuild-vb
Create a notification rule for this project
pwa-codebuild-vb
Actions
Create trigger
Edit
Clone
Debug build
Start build with overrides
Start build
Configuration
Source provider
GitHub
Primary repository
Kyra725/MyFirstRepo
Artifacts upload location
Service role
Public builds
Disabled
Build history
Batch history
Project details
Build triggers
Metrics
Debug sessions
Stop build
View artifacts
View logs
Delete builds
Retry build
Debug build

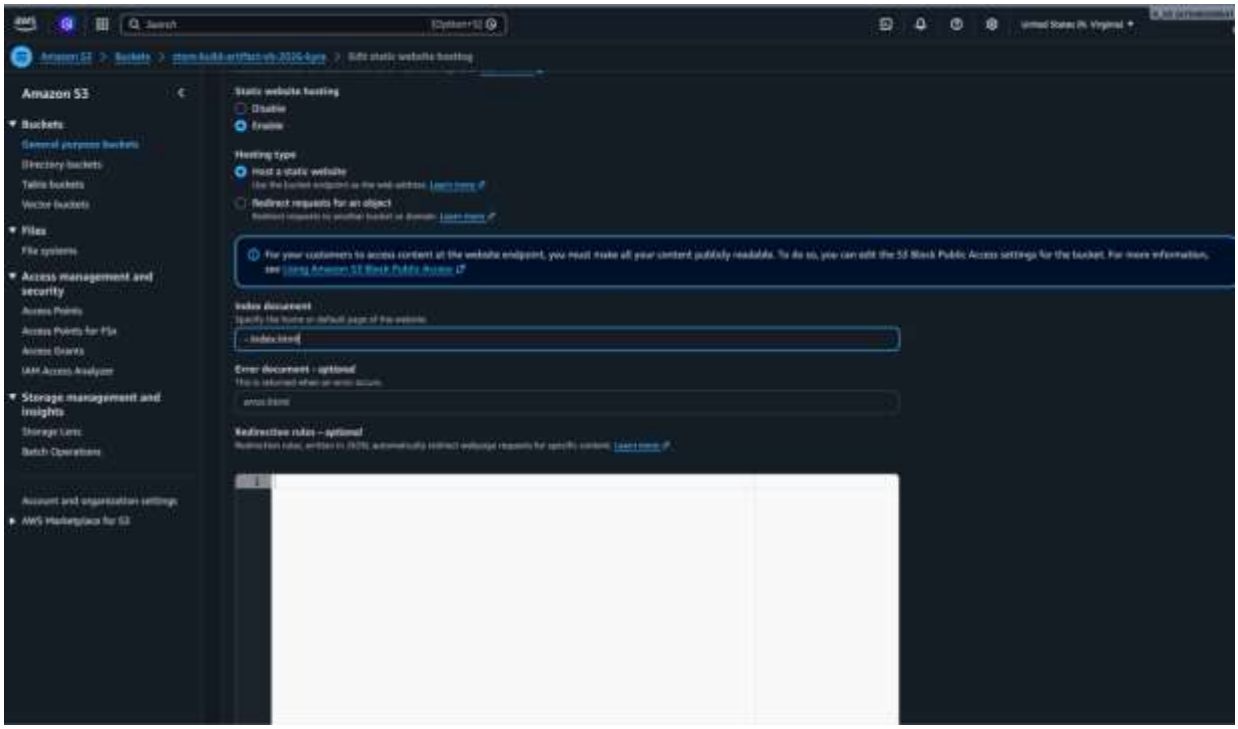

Developer Tools > CodeBuild > Build projects > Create build project
New service role
Create a service role in your account
Existing service role
Choose an existing service role from your account
Role name
codebuild-pwa-codebuild-vb-service-role
Type your service role name
Additional configuration
Timeout, privileged, certificate, VPC, compute type, environment variables, file systems, auto-retry, registry credential
Buildspec
Build specifications
Insert build commands
Store build commands as build project configuration
Use a buildspec file
Store build commands in a YAML-formatted buildspec file
Buildspec name - optional
buildspec.yml
Batch configuration
You can run a group of builds as a single execution. Batch configuration is also available in advanced option when starting build.
Define batch configuration - optional
You can also define or override batch configuration when starting a build batch.

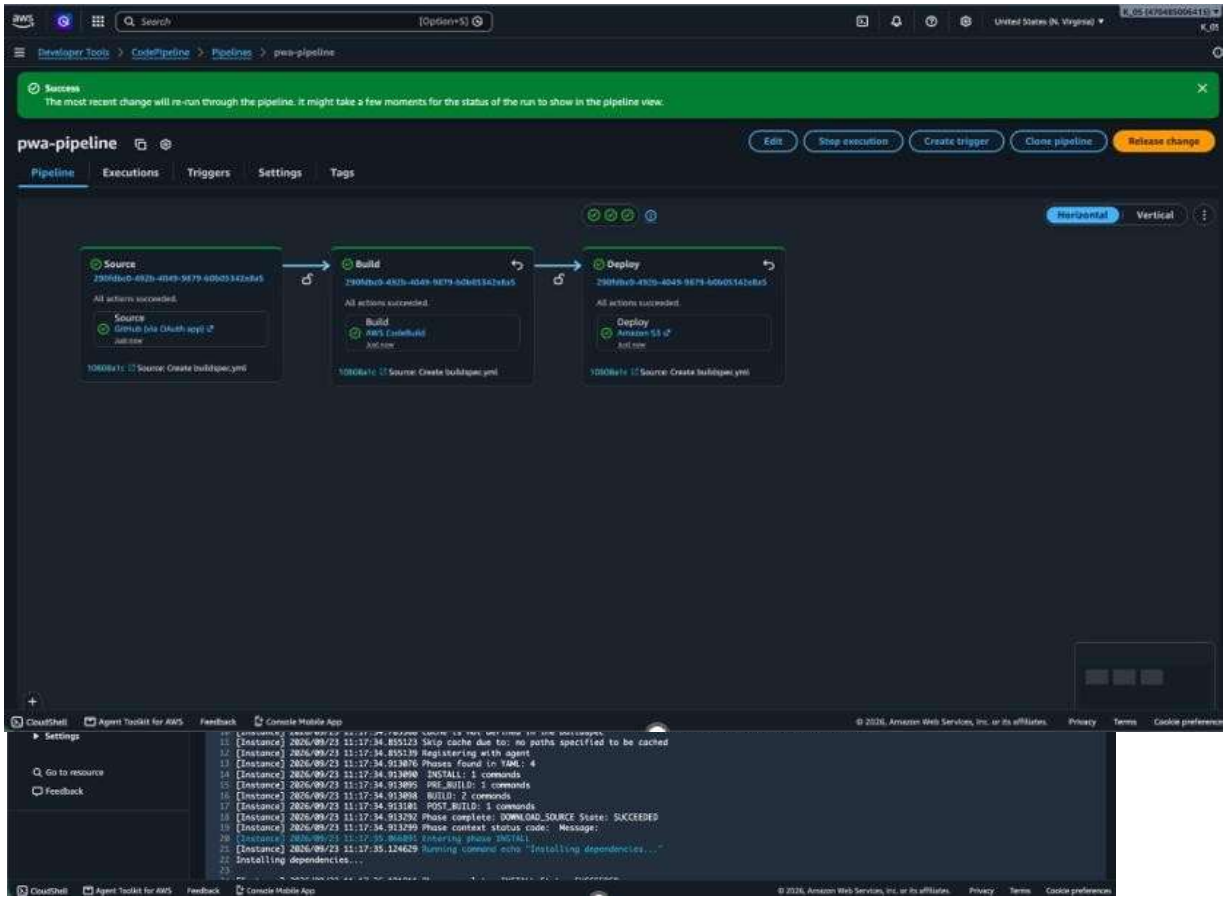

Create build project
Project configuration
Project name
pwa-codebuild-vb
Project type
Default project
Runner project
Additional configuration
Source
Add source
Source 1 - Primary

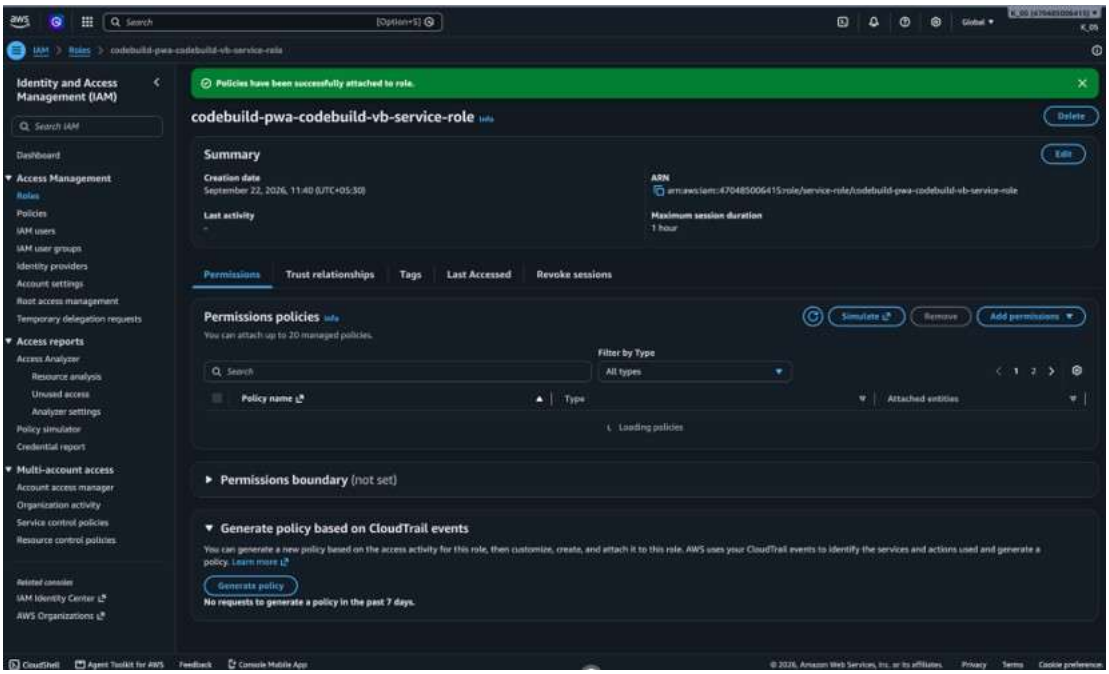

Updates Available
Use override credentials for this project only
Repository
Repository in my GitHub account
Public repository
GitHub scoped webhook
https://github.com/Kyra725/MyFirstRepo.git
Rebuild every time a code change is pushed to this repository
Build type
Single build
Batch build
Comment approval
ALL_PULL_REQUESTS
Approver roles

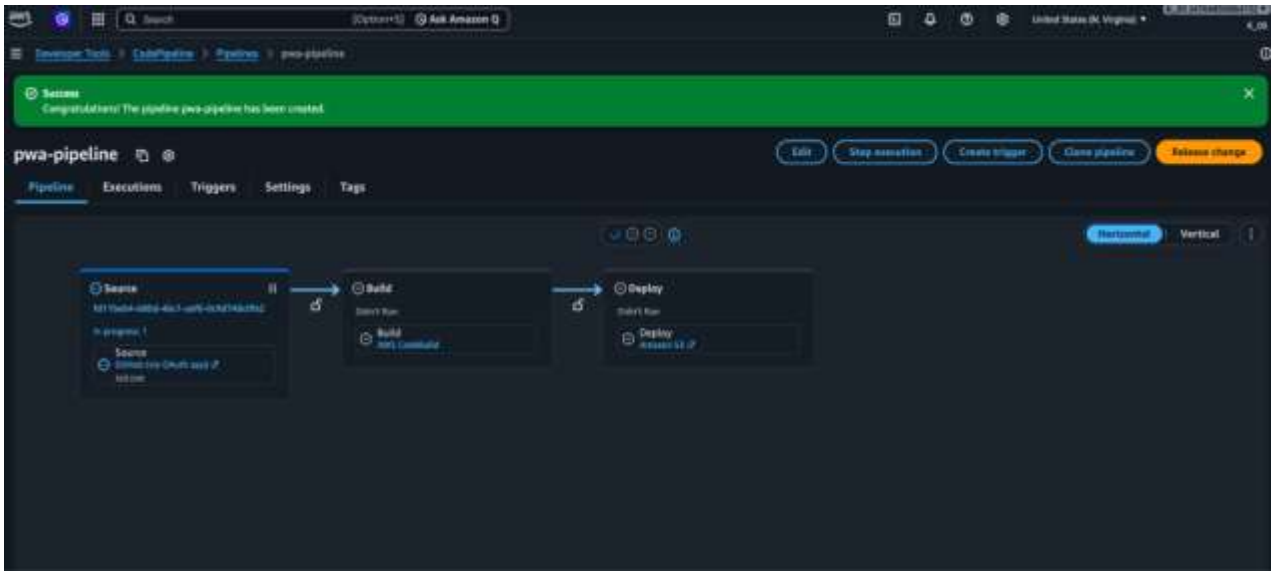

us-east-1.console.aws.amazon.com/codesuite/codepipeline/pipeline/new?region=us-east-1
Phyllotaxis - Wikip...
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Search
[Option+S]
Ask Amazon Q
Developer Tools > CodePipeline > Pipelines > Create new pipeline
Step 1
Choose creation option
Step 2
Choose pipeline settings
Step 3
Add source stage
Step 4 - optional
Add build stage
Step 5 - optional
Add test stage
Step 6
Add deploy stage
Step 7
Review
Add source stage Info
Step 3 of 7
Source
Source provider
This is where you stored your input artifacts for your pipeline. Choose the provider and then provide the connection details.
GitHub (via OAuth app)
Grant AWS CodePipeline access to your GitHub repository. This allows AWS CodePipeline to upload commits from GitHub to your pipeline.
Connected
You have successfully configured the action with the provider.
The GitHub (via OAuth app) action is not recommended
The selected action uses OAuth apps to access your GitHub repository. This is no longer the recommended method. Instead, choose the GitHub (via GitHub App) action to access your repository by creating a connection. Connections use GitHub Apps to manage authentication and can be shared with other resources. Learn more
Repository
Kyra725/MyFirstRepo
Branch
main
Enable automatic retry on stage failure
Cancel
Previous
Next

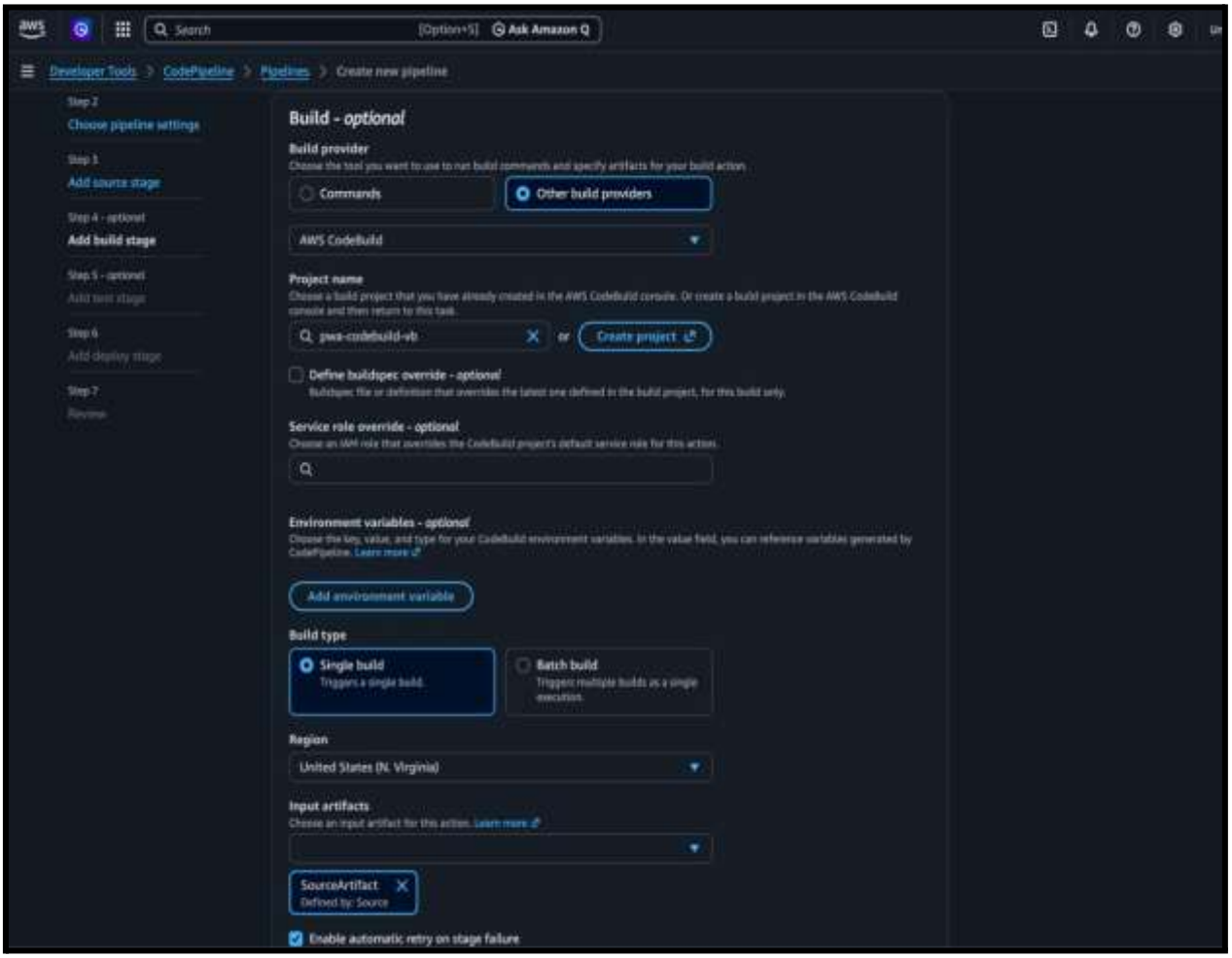

Welcome to My AWS CI/CD Website
This website is deployed automatically using AWS CodePipeline.
Source: GitHub
Build: AWS CodeBuild
Deployment: Amazon S3


Amazon S3
Static website hosting
Enable
Disable
Hosting type
Host a static website
Redirect requests for an object
Index document
Error document - optional
Redirection rules - optional


Success
The most recent change will re-run through the pipeline. It might take a few moments for the status of the run to show in the pipeline view.
pwa-pipeline
Edit
Stop execution
Create trigger
Clone pipeline
Release change
Pipeline
Executions
Triggers
Settings
Tags
Horizontal
Vertical
Source
Build
Deploy
[Instance] 2026/09/23 11:17:34.855123 Skip cache due to: no paths specified to be cached
[Instance] 2026/09/23 11:17:34.855139 Registering with agent
[Instance] 2026/09/23 11:17:34.913076 Phases found in YAML: 4
[Instance] 2026/09/23 11:17:34.913090 INSTALL: 1 commands
[Instance] 2026/09/23 11:17:34.913095 PRE_BUILD: 1 commands
[Instance] 2026/09/23 11:17:34.913098 BUILD: 2 commands
[Instance] 2026/09/23 11:17:34.913101 POST_BUILD: 1 commands
[Instance] 2026/09/23 11:17:34.913292 Phase complete: DOWNLOAD_SOURCE State: SUCCEEDED
[Instance] 2026/09/23 11:17:34.913299 Phase context status code: Message:
[Instance] 2026/09/23 11:17:35.124629 Running command echo "Installing dependencies..."
Installing dependencies...


Identity and Access Management (IAM)
Policies have been successfully attached to role.
codebuild-pwa-codebuild-vb-service-role
Delete
Summary
Edit
Creation date
September 22, 2026, 11:40 (UTC+05:30)
ARN
Last activity
Maximum session duration
1 hour
Permissions
Trust relationships
Tags
Last Accessed
Revoke sessions
Permissions policies
You can attach up to 20 managed policies.
Simulate
Remove
Add permissions
Filter by Type
All types
Policy name
Type
Attached entities
Loading policies
Permissions boundary (not set)
Generate policy based on CloudTrail events
You can generate a new policy based on the access activity for this role, then customize, create, and attach it to this role. AWS uses your CloudTrail events to identify the services and actions used and generate a policy. Learn more
Generate policy
No requests to generate a policy in the past 7 days.


Success
pwa-pipeline
Edit
Stop execution
Create trigger
Clone pipeline
Release change
Pipeline
Executions
Triggers
Settings
Tags
Horizontal
Vertical
Source
Build
Deploy



Conclusion: Thus, I successfully implemented an AWS CI/CD pipeline using GitHub, AWS CodePipeline, AWS CodeBuild, and Amazon S3. I learned how to connect a GitHub repository with AWS, create and configure a CodeBuild project using buildspec.yml, and automate the build and deployment process through CodePipeline. I also learned how to host the deployed application as a static website on Amazon S3 and how changes made to the GitHub code can be automatically reflected on the deployed website.

LO4, LO6 mapped successfully.